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ORIGINAL ARTICLE

Understanding root humping in high-power laser welding of stainless steels: a combination approach

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Abstract
High-power laser welding is an efficient way for joining the thick plates in the industries. However, full penetration laser welding of thick plates is prone to form weld imperfections such as undercuts, root humps, and porosity. In this paper, numerical simulation and laser welding experiments are combinedly carried out to investigate the dynamics of the keyhole and molten pool and understand the formation mechanism of root humping during full penetration fiber laser welding of thick stainless steel plates. Besides, the microstructure of the welded joint is analyzed. Good agreements are obtained between the numerical simulation and experimental results. The results indicate that recoil pressure, surface tension, and gravity are the main driving forces for the formation of root hump. The melt flow driven by the recoil pressure near the bottom of the keyhole is an initial and direct formation cause of the root hump. Welding speed is an important factor for the growth of root hump. Local solidification of the rear weld pool path is a hinder for the continuous growth of root hump and can cause a periodic root humping weld.

Keywords Laser welding · Keyhole · Weld pool · Root hump · Numerical simulation

1 Introduction

The 10-kW-level high-power lasers make the thick plates to be welded with single pass and hence improve the production efficiency. However, a good weld appearance is still difficult to be obtained during high-power laser welding of thick plates. One of the most common defects is root humping with complete penetration of the weld. Underfilling on the top surface always occurred with the formation of root humping and deteriorated the properties of joint [1, 2]. Unlike the hump formation on the top surface [3, 4], gravitational forces and internal melt flows have an effect on root humping [5]. As the thicker plates need to be welded with a higher laser power, root humping is becoming more problematic and requires in-depth investigation.

Many workers have studied root humping in welding processes experimentally. Kaplan and Wiklund [6] found that the welds of thick metal plates were sagged when a high line energy was used. Ilar et al. [7] studied the effect of the combination of surface tension, melt supply, and gravity effort on root humping, and there was a significant difference between the root humping and humping on the top surface. They also investigated the formation mechanism of root humping, and no correlation was found between hump size and distance due to the failure in melt flow to the root [8, 9]. Haug et al. [10] demonstrated that the penetration depth of the keyhole in laser welding was changeable, and the root humps are generated when the molten metal flowed out to form the droplets due to conductive heating behind the keyhole at partial penetration welding mode. Ohnishi et al. [11] found that hump-free joints can be obtained when a complete penetration keyhole was produced with a sufficient laser power. They also indicated that lower viscosity and less penetration of weld pool caused root humps. Pan et al. [12] observed the processes of hump formation during laser-arc hybrid welding of thick plates

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using high-speed X-ray transmission imaging system. They pointed out that the melt flows inside the weld pool in laser-arc hybrid welding had influence on the formation of root humps. Frostevarg and Haeussermann [13] observed the melt flows at the bottom surface using high-speed imaging system and indicated that melt was pushed out near the keyhole and flowed backward subsequently, which finally formed a droplet under the action of surface tension. Qi et al. [14] found that laser welding with steady electromagnetic force could minimize the input of laser energy, decrease the sensitivity to laser power for welding process, and then prevent the formation of the root humps. Zhang et al. [15] also demonstrated that root humps were formed when the heat input was increased. In our previous studies [16], a high-speed imaging system was also utilized for observing underfill and root humping processes based on modified “sandwich” method [17, 18]; it is found that a large volume root hump was generated periodically during full penetration welding at a positive defocus, and a small volume root hump was formed with instabilities of the keyhole at a negative defocus. Zhang et al. [19–21] studied designs, a method for determining the shape and size of the molten pool, and analyzed the energy distribution on keyhole wall. Besides, they also studied the forces on the surface of the molten pool.

Although many kinds of theoretical models have been proposed to explain hump formation on the top surface [5, 22–28], there are few computational models applied for investigating the humping phenomenon on the bottom surface. Petring et al. [22] presented a pressure balance equation for maximum root width at any location along the root length, after considering the downward momentum of the melt. Blecher et al. [23] developed a force balance equation between the weight of the liquid metal and the surface tension to explain the competing forces driving the formation of root hump. Bachmann et al. [24, 25] proposed a three-dimensional laminar steady-state numerical model to calculate the velocity, pressure, temperature, and electromagnetic field distributions during full penetration laser welding of thick aluminum and stainless steel plates with an electromagnetic weld pool support system. Guo et al. [26] developed a computational fluid dynamic model and made a comparative investigation of the dynamic forces on the weld pool in flat and horizontal position. They found that it was difficult to obtain a good weld without applying a weld pool supporting system [26]. Frostevarg [5] reviewed the previous researches and presented an analytical model to reveal the role of the surface tension during the formation of root humping.

In conclusion, root hump is a most problematic imperfect [23], and there are insufficient in-depth studies on the root humping mechanism in high-power laser. Therefore, this paper intends to figure out the formation mechanism of root humping in autogenous laser welding, where the welding is operated on thick stainless steel plates with the use of a 10-kW fiber laser. With the purpose to investigate root hump in details, a three-dimensional comprehensive model was established to simulate the keyhole and weld pool profile characteristics when the root humps were generated. In the model, transient heat transfer, fluid flow, and keyhole dynamics were taken into account. The high-speed photography was utilized for observing the keyhole and weld pool dynamics during laser welding of a modified “sandwich” specimen or a 12-mm-thick stainless steel plate. Afterward, the paper focused on the discussion on the formation mechanisms of the root humps. Last but not the least, the microstructure of the butt joint was investigated for verifying the formation mechanisms of the root humps.

2 Material and methods

2.1 Experimental procedures

The setup of experiments is shown in Fig. 1. A continuous wave fiber laser (IPG YLS–10000) is chosen as the laser source, whose main specifications are as follows: (1) the options of beam parameter product are 3.8, 5.2, and 7.0 (mm × mrad); (2) the maximized average power is 10 kW; and the output fiber core diameters include 100, 150, and 200 μm. The laser beam emitted from the end of the optical fiber is collimated by a lens with a focal length of 150 mm, which is then focused on the specimen surface by another lens with a focal length of 300 mm.

Figure 2 illustrates the schematic representation of the experimental setup with different camera configurations. Type 304 austenitic stainless steel is selected as the materials to be welded, whose thickness is 12 mm. Table 1 lists the plate materials’ chemical composition. The experimental parameters are listed in Table 2. The shielding gas was nitrogen supplied from the side during bead-on-plate welding.

2.2 Numerical simulation

With the aim to figure out the root humping’s formation mechanisms and the weld pool dynamics in high-power laser welding, the welding process was numerically simulated with the 3D laser welding model developed by our team [27]. In the model, transient heat transfer, fluid flow, and keyhole dynamics were taken into account. The crucial physical factors influencing the keyhole dynamics, such as the recoil pressure, Marangoni shear stress, and surface tension, were accurately incorporated with sharp interface method [29–31]. For conveniently characterizing the absorption of the high-power laser beam energy absorbed by the keyhole wall, a ray tracing algorithm was adopted by following these related studies [31, 32].

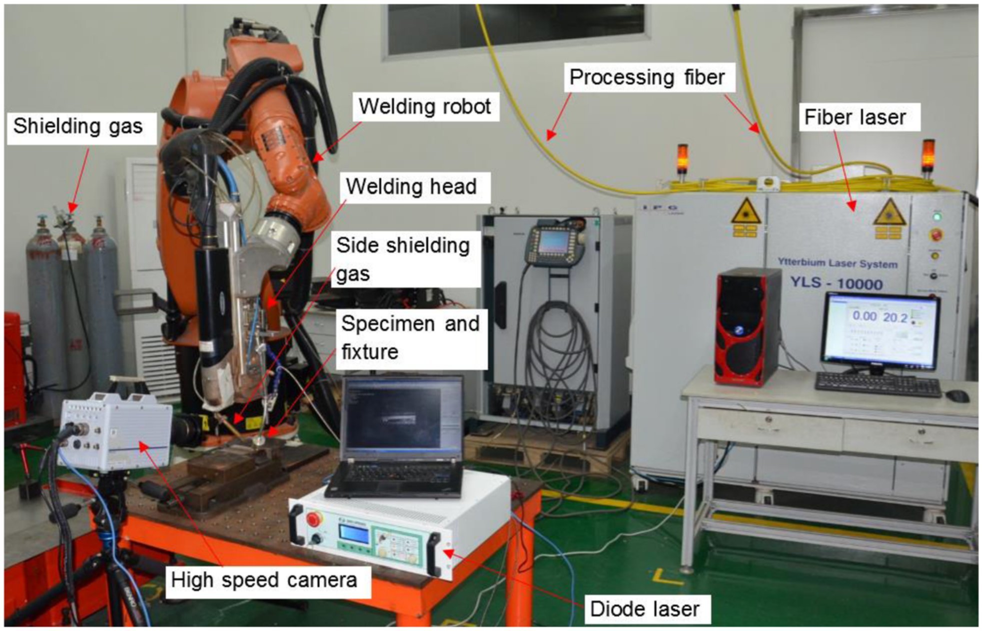

Fig. 1 Experimental setup

In this model, we assume that the molten metal liquid is an incompressible fluid [33] and uses the mixture model to describe the flow heat transfer process:

$$\nabla \cdot \overrightarrow{U} = 0 \tag{1}$$

$$\rho\left(\frac{\partial \overrightarrow{U}}{\partial t} + \left(\overrightarrow{U} \cdot \nabla\right)\overrightarrow{U}\right) = \nabla \cdot \left(\mu \nabla \overrightarrow{U}\right) - \nabla p - \frac{\mu}{K}\overrightarrow{U} - \frac{C\rho}{\sqrt{K}}\left|\overrightarrow{U}\right|\overrightarrow{U} - \rho \overrightarrow{g}\beta(T - T_{ref}) \tag{2}$$

$$\rho C_p\left(\frac{\partial T}{\partial t} + \left(\overrightarrow{U} \cdot \nabla\right)T\right) = \nabla \cdot (k\nabla T) + Q \tag{3}$$

where $\overrightarrow{U}$ is the three-dimensional velocity vector, ρ is the density, p is the pressure, μ is the viscosity, $\overrightarrow{g}$ is the three-dimensional gravitational acceleration vector, β is the thermal expansion coefficient, T_{ref} is the reference temperature, and K is the Carman-Kozeny coefficient in the mixed phase model. C_p is the specific heat and k is the thermal conductivity [34, 35].

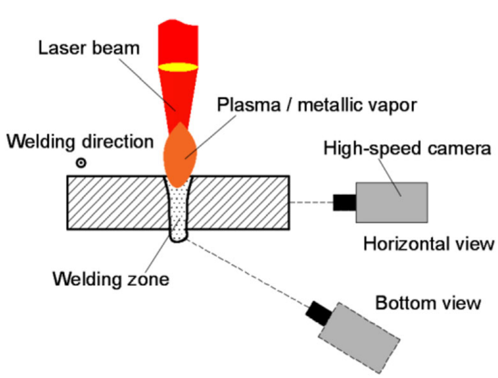

Fig. 2 Schematic representation of the experimental setup with different camera configurations

Level set method is adopted to track the keyhole-free surface evolutions, and the time-dependent keyhole profiles is described as [36]:

$$\frac{\partial \phi}{\partial t} + \overrightarrow{U} \cdot \nabla \phi = 0 \tag{4}$$

where ϕ is a signed distance. The transient free surface of the keyhole can be described as the zero contour of the distance field ϕ. $\overrightarrow{U} = (u, v, w)$ is the velocity of the weld pool near the free surface.

For the fluid flow boundary condition of the keyhole and weld pool, sharp interface method is adopted [29–31]. On the tangential direction of the free interface, the viscous stress momentum caused by the Marangoni forces and viscous forces of the molten metal is calculated with a derived tensor form. The surface tension, back pressure, and fluid pressure on the normal direction of the keyhole free interface can be described as:

$$p_f = p_r + \sigma\kappa + 2\mu \overrightarrow{n} \cdot \nabla \overrightarrow{U} \cdot \overrightarrow{n} \tag{5}$$

where the subscript f indicates the free interface, σ is the surface tension coefficient, κ is the curvature, and μ is the density of molten metal. A recent improved surface pressure model is utilized for calculating p_r here [29, 37], which can be expressed as:

Table 1 Chemical composition of the used sample material

Element	C	Mn	P	S	Si	Cr	Ni	N	Fe
(Wt.%)	0.07	2.00	0.045	0.03	0.075	18.28	8.15	0.1	Balance

Table 2 Welding parameters used in the experiments

Parameter	Value
Laser power (W)	10,000
Weld speed (m/min)	0.9, 1.2
Defocus (mm)	+10
Shielding gas type	N_2
Shielding gas flow (l/min)	20, 30
Weld sample	Modified sandwich, bead-on-plate

$$P_r = \begin{cases} P_{amb} & 0 \le T_{fs} < T_L \\ \frac{1+\beta_r}{2} P_0 \exp\left(\frac{\Delta H_v}{k_B T_v}\left(1 - \frac{T_v}{T_{fs}}\right)\right) & +\infty > T_{fs} \ge T_R, \\ P_c(T_{fs}) & T_L \le T_{fs} < T_R \end{cases} \quad (6)$$

where P_{amb} is the ambient pressure, K_B is the Boltzmann constant, T_v is the boiling point, P_0 is the atmospheric pressure, T_{fs} is the keyhole surface temperature, p_c (T_{fs}) is a smooth cubic interpolation curve, and T_L and T_R are the temperatures of the two tangent points between the smooth curve p_c (T_{fs}) with the ambient line and the recoil pressure curve. The parameters of the surface pressure model are cited from [37].

In regard to the thermodynamics boundary condition of the keyhole and weld pool, a robust ray tracing algorithm is adopted to calculate the multiple reflections of the laser beam on the metal surface [31]. The laser beam energy distribution is modeled as a Gaussian function:

$$I_0(r,z) = 3P/\left(\pi R^2\right)\exp\left(-3\left(r^2\right)/R^2\right) \quad (7)$$

where R is the effective beam radius and P is the laser power.

According to the ray tracing algorithm, the laser beam is split into many bundles at first. The bundle has its direction ($\overrightarrow{I}_0$) and starting point coordinate. The energy of each bundle is absorbed based on the following Fresnel absorption formula [32], and the Fresnel absorption coefficient is:

$$\alpha_{Fr}(\theta) = 1 - \frac{1}{2}\left(\frac{1+(1-\varepsilon\cos\theta)^2}{1+(1+\varepsilon\cos\theta)^2} + \frac{\varepsilon^2 - 2\varepsilon\cos\theta + 2\cos^2\theta}{\varepsilon^2 + 2\varepsilon\cos\theta + 2\cos^2\theta}\right) \quad (8)$$

where θ is the angle between the incident beam and the normal vector of surface cell and ε is a coefficient related to the types of lasers. From above, the heat flux on metal surface from laser beam irradiation can be determined by:

$$q = I_0(r,z)\left(\overrightarrow{I}_0 \cdot \overrightarrow{n}\right)\alpha_{Fr}(\theta_0) + \sum_{n=1}^{m} I_n(r,z)\left(\overrightarrow{I}_n \cdot \overrightarrow{n}\right)\alpha_{Fr}(\theta_n) \quad (9)$$

Technically, it is equally important to accurately discretize the equations and to develop efficient solvers during the high-power laser welding simulation. Here, the higher-order finite difference method was used to discretize the equations above. The transient term, convection term and diffusion term were discretized in the third-order Runge-Kutta scheme, the fifth-order WENO scheme [38], and the second-order central difference scheme, respectively. A projection method was adopted to solve the momentum conservation equation, with a semi-implicit method to discretize the energy conservation equation [39]. The higher-order explicit method was also adopted to discretize the LS equation [38]. The computer program was developed with the C++ programming language. Besides, the OpenMP language was applied in parallel computation. The simulations took about 70 h to complete 500-ms laser welding process by using our in-house procedure in parallel on a small workstation (Intel Xeon E5, 2.20 GHz CPU, 16 cores). Table 3 lists the physical parameters used in the simulation, while Fig. 3 reveals the detailed calculation flow chart.

3 Results and discussion

3.1 Predication of root hump formation

In order to study the formation of root hump in stainless steel high-power laser welding, the welding process under 10-kW condition is simulated, and the corresponding fluid flow prediction results of keyhole and weld pool are shown in Fig. 4. It can be seen that the formation of the root hump is closely related to the flow behavior of the weld pool during penetration laser welding. There is a typical flow during the welding process, that is, the molten metal in the front wall of the keyhole mainly flows downward [31]. The average flow velocity is close to 3 m/s, and the maximum value is up to 10 m/s. In the process of non-penetration, when this part of the molten metal reaches the bottom of the keyhole, due to the blockage of unmelted solid below the bottom of the keyhole, it changes to the rear of the weld pool (opposite to the welding direction), and no hump is formed, as shown in the Fig. 4a. When the plate is melted, this downward flowing metal liquid impacts

Table 3 Physical parameters used in the simulations

Physical parameters	Units	Value
Density ρ	$kg \cdot m^{-3}$	7200
Thermal conductivity coefficient k	$W \cdot m^{-1} \cdot K^{-1}$	35 (298 K)
Specific heat C_p	$J \cdot kg^{-1} \cdot K^{-1}$	760 (298 K)
Kinematic viscosity μ	$kg \cdot m^{-1} \cdot s^{-1}$	0.006
Latent heat of fusion C_f	$J \cdot kg^{-1}$	6.0×10^5
Latent heat of evaporation C_v	$J \cdot kg^{-1}$	6.52×10^6
Solidus temperature T_s	K	1727
Liquids temperature T_l	K	1697
Evaporation temperature T_v	K	3100
Surface tension coefficient σ	$N \cdot m^{-1}$	1.0
Thermal-capillary force coefficient σ_T	$N \cdot m^{-1} \cdot K^{-1}$	-0.43×10^{-3}

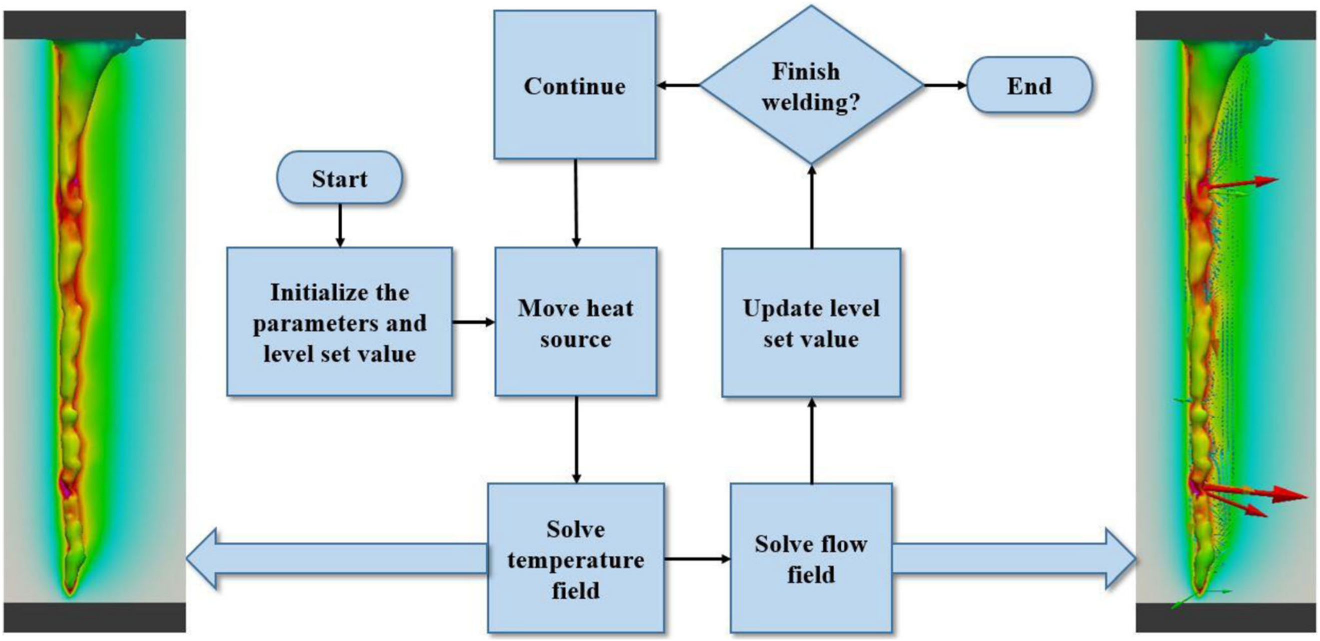

Fig. 3 The simulation flow chart of high-power laser welding

the molten metal at the bottom. When this impact is stronger than the surface tension and viscous action of the bottom weld pool, it is observed that the bottom bulge begins to appear, as revealed in Figs. 4b–d. At the beginning, the metallic vapor impinges on the rear wall of keyhole to form a local expansion, and the metal liquid near the bottom of the keyhole flows downward to form a bulge, as shown in Fig. 4b. Subsequently, the expansion area moves downward, and the bottom of the keyhole opens. The metal liquid continues to flow downward, and the droplet at the bottom of the keyhole increases, as shown in Fig. 4c. Finally, the metallic vapor wave moves upward; the back wall of the bottom of the keyhole is compressed; the keyhole is closed; and the hump begins to form, as shown in Fig. 4d. Therefore, it can be considered that the flow of molten metal near the bottom of the keyhole is a direct cause of the occurrence of the hump in the penetration state.

In order to further figure out the formation mechanism of high-power laser welding hump, the dynamic behavior of the

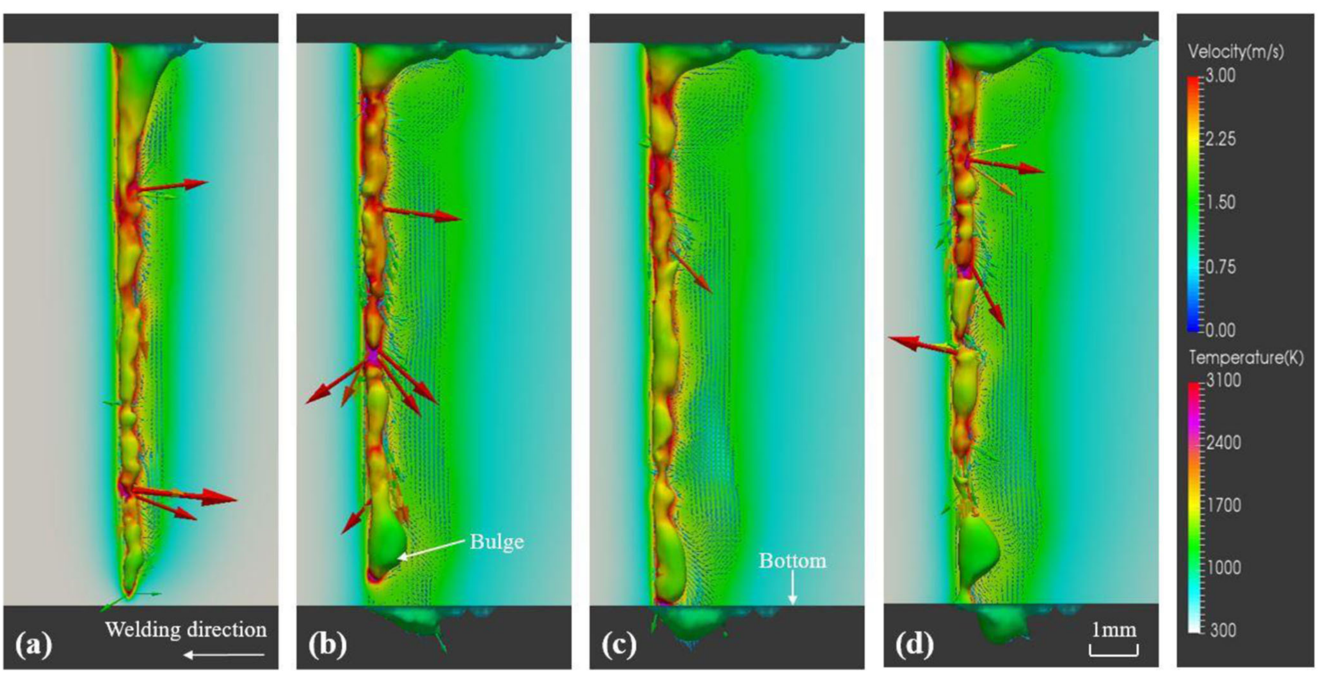

Fig. 4 The formation of root humping. **a** t = 87.978 ms. **b** t = 250.291 ms. **c** t = 251.284 ms. **d** t = 253.888 ms

bottom of the keyhole during the formation of hump is studied in detail. As can be seen from Fig. 5, the grown hump is behind the bottom of the keyhole. It can be inferred that the growth of the hump is closely related to the behavior of the bottom weld pool. With the progress of the welding process, the metal liquid at the bottom of the weld pool continues to aggregate, which finally forms a droplet, as shown in Fig. 5a. When the recoil pressure and gravity are stronger than the surface tension, the droplet forms a spatter, as shown in Fig. 5b. From the flow trend, the metal liquid at the bottom of the weld pool mainly flows downward toward the rear, mainly because the surface tension changes the direction of the downward metal liquid at the bottom of the keyhole, demonstrated in Fig. 5c. This flowing metal liquid encounters the obstruction of the solidification front, and the flow slows down and accumulates so that the hump gradually grows up, as shown in Fig. 5d. It is generally known that when the weld enters the quasi-steady state, the weld pool behind the keyhole is relatively large and can be maintained in the melting stage for a long time. The corresponding gravity is about 0.124 N, and the surface tension at the bottom is estimated to be 0.188 N, which increases as the hump volume increases. Therefore, the effect of gravity on the growth of the root humping is not negligible. In summary, gravity and recoil pressure can promote the formation of hump. It can be considered that the downward flow of the bottom of the keyhole is the main factor for the growth of the hump. Moreover, the surface tension and the advancement speed of the solidification front of the weld pool can affect the size of the hump. The advancement speed of the solidification front of the weld pool is related to the welding speed.

The fluid flow and its mechanism in the corresponding weld pool are also analyzed in detail. Figure 6 displays the temperature and velocity characteristics of the metal liquid on the surface of the bottom weld pool during the laser welding hump formation. Since the laser mainly acts inside the keyhole, the temperature of the bottom weld pool is much lower than the internal temperature of the keyhole, which is about 1500 K. When the hump begins to form, the molten metal in the bottom weld pool is subjected to gravity and recoil pressure and flows downward to form a bulge. The flow direction is mainly vertically downward along the keyhole, as shown in Fig. 6a. When the droplet grows to a certain size, the surface tension of the raised droplet increases, and the flow direction of the metal liquid on the surface of the weld pool becomes toward the weld pool, as shown in Fig. 6b.

After a period of time, the back of the weld pool is cooled, and the passage of the molten metal to the backward flow is gradually cut off. The flow velocity of the surface of the molten droplet is lowered at the rear of the weld pool, and the metal liquid in the front part of the weld pool forms a bulge again, as shown in Fig. 6c. The molten metal accumulates in

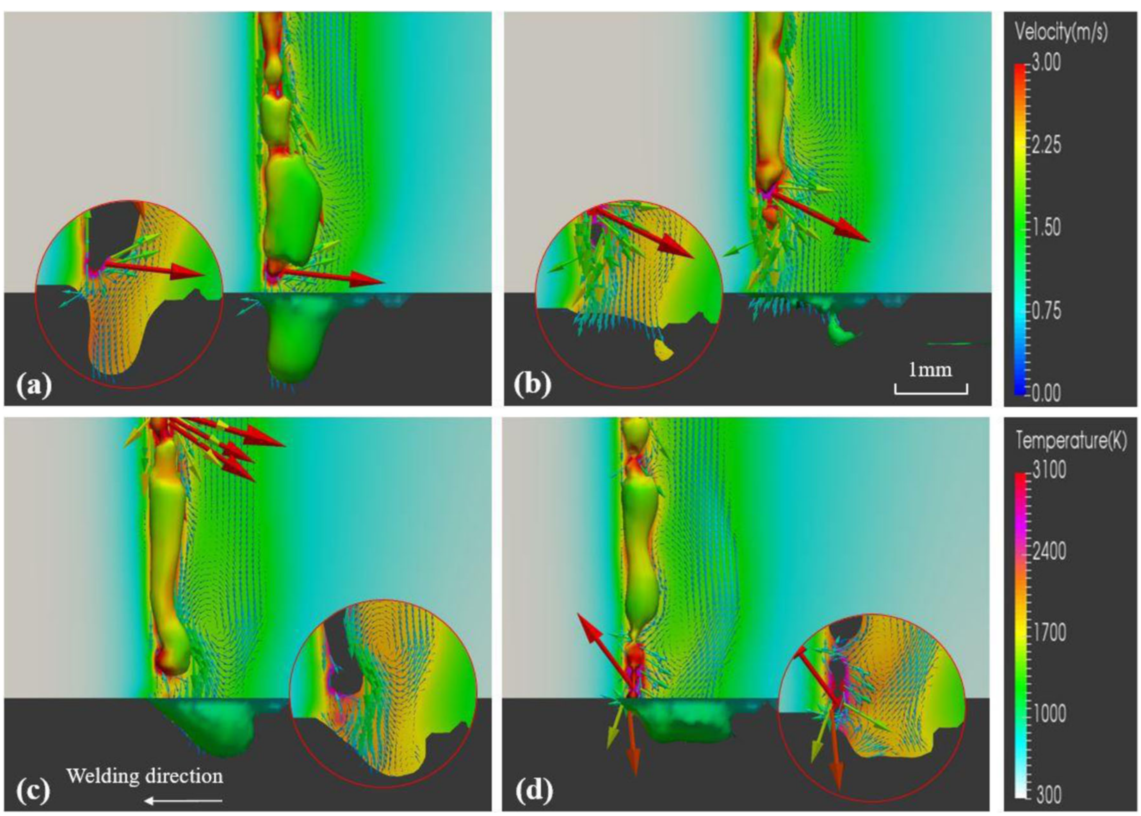

Fig. 5 The behavior of keyhole when the root humping is formed. **a** t = 233.762 ms. **b** t = 239.734 ms. **c** t = 263.156 ms. **d** t = 266.064 ms

the back of the weld pool, and the hump begins to solidify. Considering that the solidification front is continuously advancing, the size of the hump is considered to be affected by the advancement speed of the solidification front, that is, the welding speed. The bulge in the front of the weld pool continues to grow, and a new hump begins to form, as shown in Fig. 6d. These velocity characteristics further confirm that the hump formation process is affected by various factors such as gravity, recoil pressure, surface tension, and the advancement speed of the solidification front of the weld pool.

3.2 Observation of root humping formation

Figure 7 shows the high-speed photographic images of the dynamic process of the keyholes and weld pools when the bottom hump is produced based on the improved "sandwich" method. When the keyhole is not penetrated, the evaporation effect at the bottom of the keyhole is severe. It can be seen that the metal vapor impacts the keyhole wall and forms a partial expansion, and the bottom weld pool is pressed toward the front edge of the keyhole to close the keyhole, as shown in Fig. 7a. As the welding process progresses, the thickness of the droplet increases below the bottom of the keyhole (Fig. 7b). As the laser energy at the bottom of the keyhole continues to increase, the evaporation effect of the metal vapor at the bottom of the keyhole is extremely severe. The backward wall of the keyhole forms a large expansion zone, while the bottom of the keyhole opens, as shown in Fig. 7c. Then, the metal vapor wave moves upward along the wall of the keyhole, the bottom wall of the keyhole is compressed, and finally the keyhole is closed, as shown in Fig. 7d. It can be seen that the interaction between the keyhole and molten pool is violent. Generally, the keyhole exit is opened discontinuously [40].

Figure 8 shows the high-speed photographic images of the dynamic change process of the bottom weld pool and metal vapor when the root humping produces. The molten metal at the bottom continuously flows backward, gathers to the trailing edge of the bottom weld pool to form a droplet (Fig. 8a), and the droplet continues to grow (Fig. 8b). As the welding process progresses, the bottom molten metal continuously collects and separates from the weld pool to form an inclined metal liquid column. The end droplet separates from the metal liquid column to form a spatter (Fig. 8c). At the same time, the metal liquid column continuously lengthens and thickens, and a new spatter is formed (Fig. 8d). As can be seen from Fig. 8d–h, the inclination angles α_1, α_2, and α_3 are constantly increasing, and the length of the bottom weld pool is also increasing (Fig. 8g and h). It can be seen from Fig. 8 that the molten melt flows downward vertically at the beginning of fully penetration of workpiece, which is attributed to the action of recoil pressure and gravity. As the welding

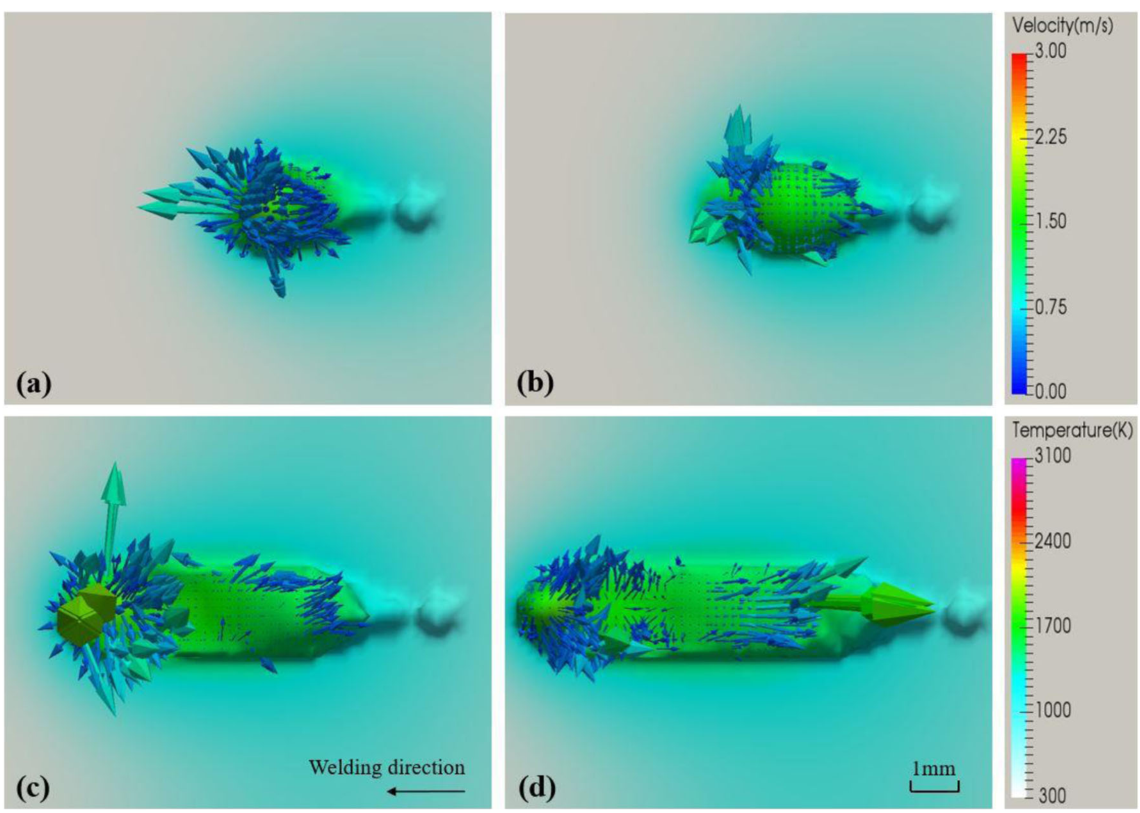

Fig. 6 The behavior of bottom weld pool when the root humping is formed. **a** t = 233.762 ms. **b** t = 264.281 ms. **c** t = 384.448 ms. **d** t = 457.008 ms

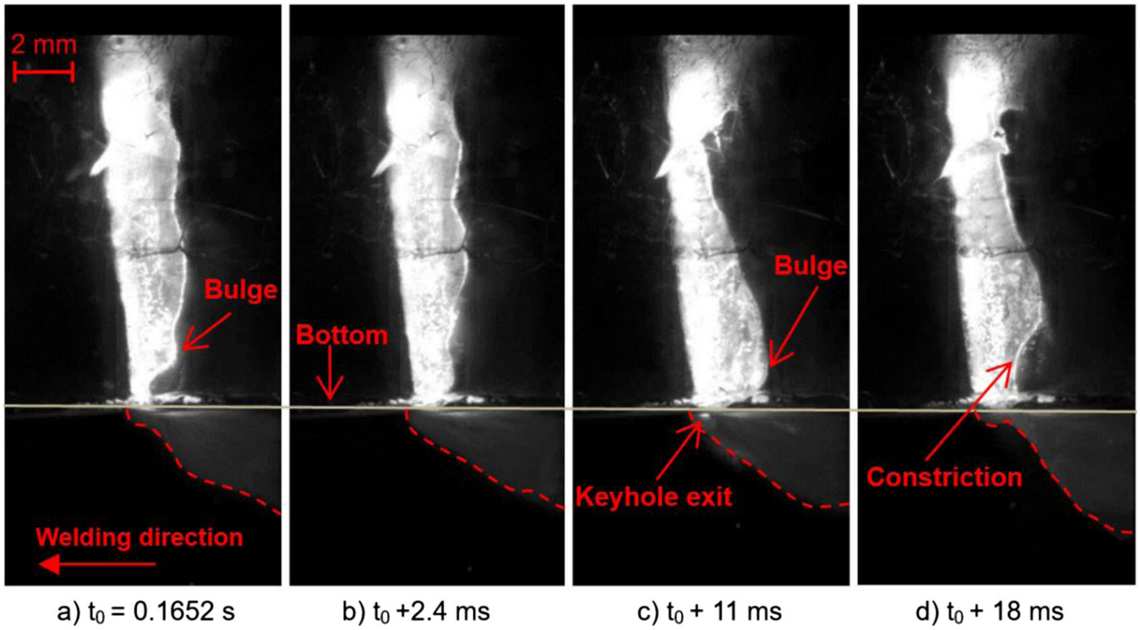

Fig. 7 Images of the longitudinal keyhole and vapor plume with the formation of root hump ($p = 10{,}000$ W, $v = 1.2$ m/min, $\Delta = +10$ mm)

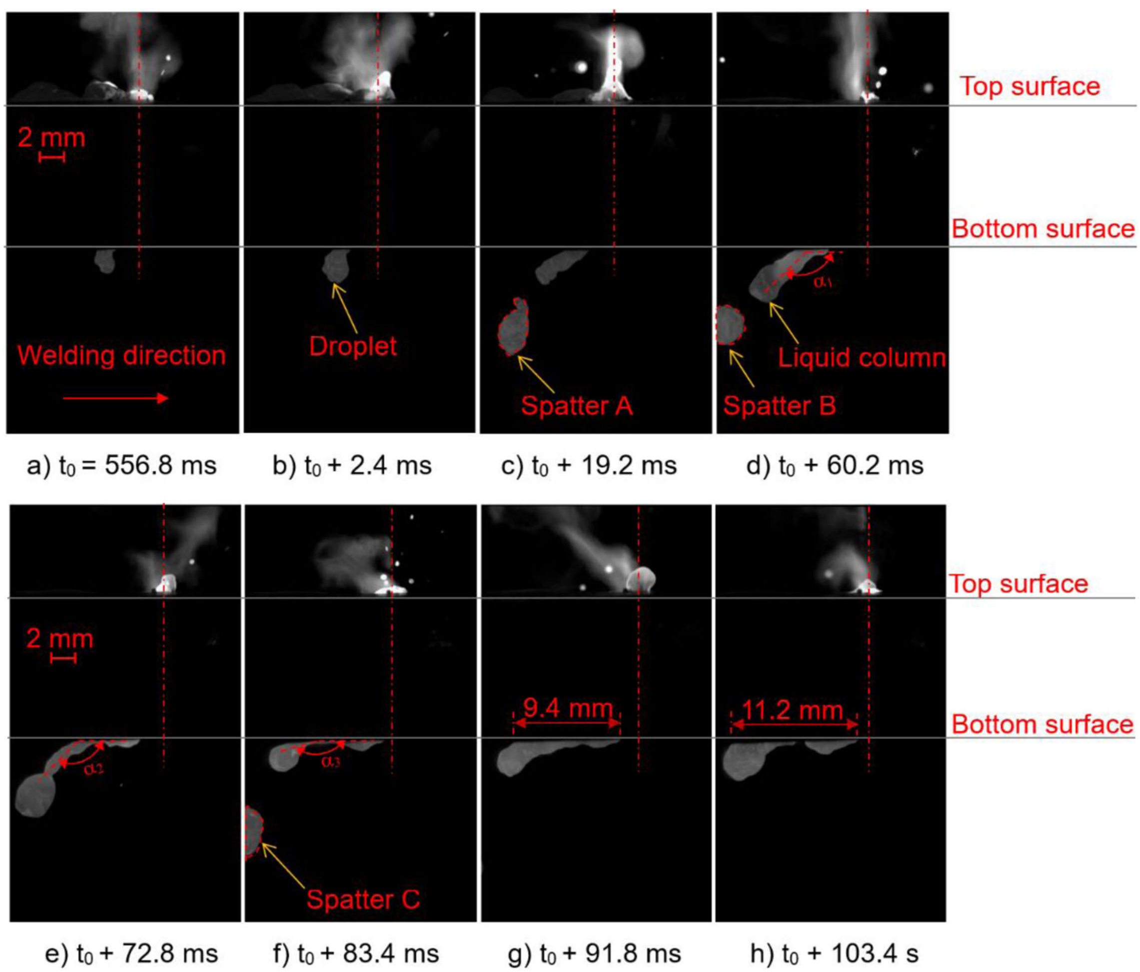

Fig. 8 High-speed images from horizontally placed camera for formation of root sags at a defocus of + 10 mm ($p = 10{,}000$ W, $v = 0.9$ m/min, $q = 20$ l/min (N_2))

process progresses, the liquid column gradually inclines and attaches to the bottom of the base metal at the action of surface tension and inertial flow.

Figure 9 shows the dynamic change process of the formation of the root humping. When the solidification front of the weld pool is not formed, the molten metal flows to the back edge of the weld pool (Fig. 9a). When the solidification front of the weld pool is formed, since the molten metal flowing backward is hindered by the solidification front, the flow velocity slows down, and the molten droplet forms under the action of gravity (Fig. 9b). At this time, as the welding speed is greater than the advancement speed of the solidification front of the weld pool, and the inside of the rear weld pool has not been solidified, the molten metal is less affected by the hindrance of the solidification front of the weld pool. Therefore, the molten metal is pulled back to the molten pool, and no droplets are formed, as shown in Figs. 9c and d. As the welding process progresses, the inside of the rear weld pool has been solidified, and the passage of the molten metal to the rear flow is cut off. Therefore, the metal liquid gradually accumulates, and the hump gradually grows up, as shown in Figs. 9e and f. From the above, the solidification of the rear weld pool path is a hinder for the continuous growth of root hump.

3.3 Mechanism understanding of root humping

In order to clearly explain the formation mechanism of the root humping, the formation mechanism of the first hump and the second and the subsequent hump is analyzed, according to the simulation analysis and experimental phenomena above. During the formation of the first root humping, the penetration state of the keyholes is from infusible to fully penetrated. When the keyhole is not penetrated, the molten metal protrudes from the rear of the bottom weld pool to form a lower liquid column, and the liquid column is vertically downward under the action of recoil pressure and gravity. Due to the surface tension and inertial flow, the liquid column gradually inclines and attaches to the bottom of the base metal. It gradually forms a hemispherical shape under the surface tension and finally adheres to the bottom of the base material to form a droplet. At this time, the welding front is completely penetrated, and the recoil pressure drives the downward flowing molten metal to flow backward under the action of surface tension, providing material for the growth of the droplets falling behind the weld pool. As the welding process progresses, the falling droplet gradually grows up. When the melt channel of the bottom weld pool reaches a certain length, the melt channel begins to solidify locally, and the channel of the molten metal flowing backward is cut off. The first root humping forms and gradually solidifies. Subsequently, the flowing backward molten metal of the welding front accumulates at the leading edge of the solidification to form a new falling droplet. The molten metal continues to flow backward at the welding front, and the falling droplet continues to grow. Afterward, a new local solidification forms, the flow passage is cut, and a new hump forms. This cycle eventually forms a periodic root humping weld.

In summary, the main driving forces for the formation of the root humping are as follows: the first is the recoil pressure. Under the driving force of the recoil pressure, the molten metal at the front of the keyhole flows down at a high speed, providing a large amount of metal materials for the formation of the root humping. The second is surface tension. On one hand, when the metal liquid column is formed, the surface tension tends to pull the metal liquid column back to the weld pool; on the other hand, the surface tension of the weld front edge weld pool tends to divert the recoil pressure driven

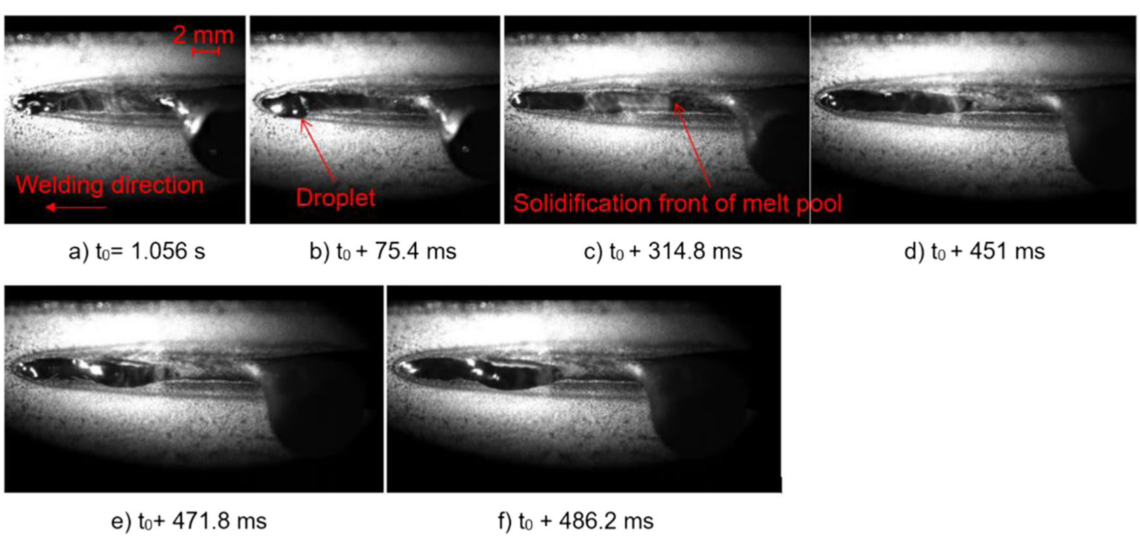

Fig. 9 Images of the bottom weld pool dynamics with formation of root sag viewed from bottom up (p = 10,000 W, v = 0.9 m/min, Δ = +10 mm, q = 30 l/min (N_2))

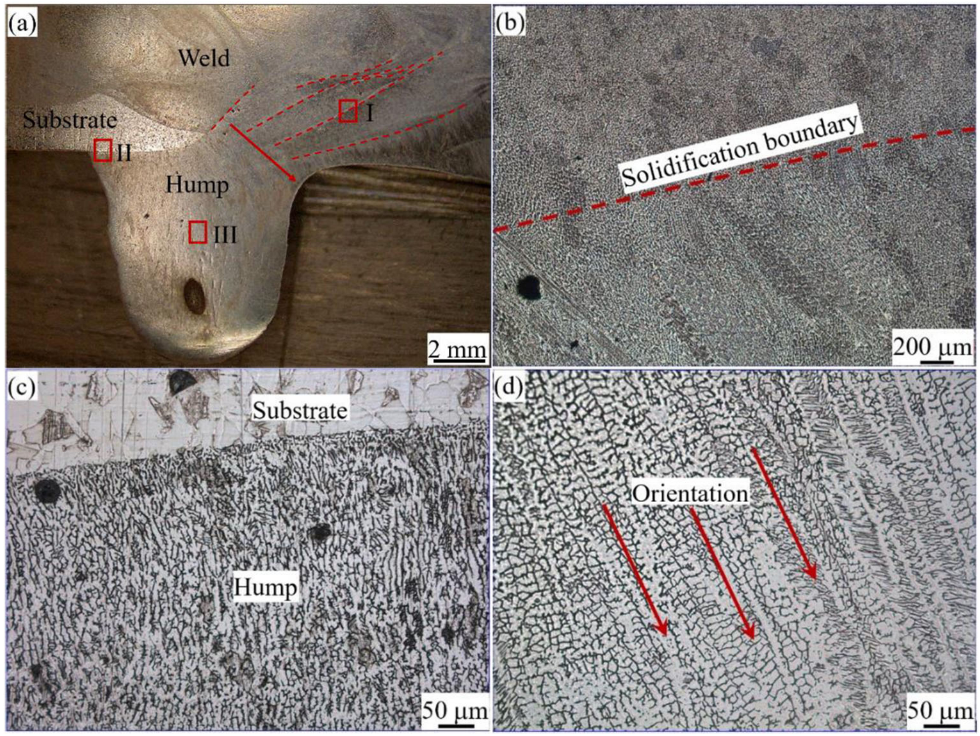

Fig. 10 Microstructure of the joint. **a** Optical micrograph of the longitudinal section of the joint. **b** Enlarged micrograph of the zone inside solid rectangle I in (**a**). **c** enlarged micrograph of the zone inside solid rectangle II in (**a**) and **d** enlarged micrograph of the zone inside solid rectangle III in (a)

downward flow to backward flow. The third is gravity. The influence of gravity is not negligible during the formation of the root humping. Because of the existence of gravity, the falling droplet cannot be easily pulled back to the weld pool by the surface tension but gradually increases, eventually forming a root humping.

Figure 10 illustrates the microstructure of the welded joint of the root hump. There are several visible solidification boundary lines ahead the root hump, as shown in Fig. 10a and b. It is reasonable to deduce that the solidification boundary line moves closer to the bottom of the weld during laser welding process. The variation trend of the solidification boundary line indicates the layer of the molten metal which flows backward at the bottom of the weld pool is changed from thick to thin. Finally, the molten metal filling the root hump is solidified, and the melt flow path at the bottom is cut off, as shown in Fig. 9c. Therefore, the root hump cannot grow continually. The wetting effect between the melt in the droplet and the base metal is excellent due to the same composition. As a result, the droplet with high temperature will flow backward against the bottom surface of the substrate. The base metal is partially melted, and the metallurgical bond between the base metal and the droplet is acquired, as shown in Fig. 10c. The typical columnar dendrites are formed at the center of the root hump and generated from the top to the bottom, as indicated with the red arrows in Fig. 10d. Especially, the boundaries among the columnar dendrites are obvious and appear to be flow. The metallographic morphology at the center of the root hump indicates that the molten metal in the root hump is constantly flowing before solidification. Based on the weld microstructure morphologies, we deduce that continuous supply of melt from welding zone toward the sagging droplet is a key forming condition of the root hump.

4 Conclusion

Motivated by the scarcity of analysis and research of the formation mechanism during high-power fiber laser welding of thick plates, we have conducted a combination approach of numerical simulation and laser welding experiments to analyze the keyhole and weld pool dynamics and to reveal the formation mechanisms of the root humps. The conclusions can be summarized as follows:

(1) Simulation and experimental results indicate that the interaction between the keyhole and molten pool is violent. The melt flow driven by the recoil pressure near the bottom of the keyhole is a direct cause of the occurrence of the root hump in the penetration state.
(2) Surface tension is one of the main driving forces for the formation of the root hump. Meanwhile, surface tension and the advancement speed of the solidification front of the weld pool at the bottom affect the size of the root humps. The advancement speed of the solidification front of the weld pool is varied with the welding speed.

(3) At the beginning of fully penetration of workpiece, the molten melt flows downward vertically at the action of recoil pressure and gravity. And then the liquid column gradually inclines and attaches to the bottom of the base metal at the action of surface tension and inertial flow. As a result, the first root hump is generated.
(4) Local solidification of the rear weld pool path causes the cutoff of the backward flow passage, resulting in fixation of the growing root hump and appearance of a new one. It is a hinder for the continuous growth of root hump and can cause a periodic root humping weld.

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